

# Tracking Events Through a News Stream: A Filter That Works, an Attribution Layer That Does Not

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*Revision of 2026-08-19. Companion to `PAPER_0_FOUNDATION.md` (the extraction substrate) and `JOINT_IE_SCALING.md` (the within-document instantiation of the same thesis). The build record — codebase attachment points, CLI, training plan, decision log, generator specification, test protocol, and the negative-sampling implementation notes — is kept verbatim in `EKF_MHT_BUILD_RECORD.md`.*

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## Abstract

A real-world event reported across a stream of news documents has a state that evolves: a death toll rises, gets revised downward, and is reported by sources that disagree, censor ("at least 12"), and lag. We build the full pipeline — synthetic stream generator, LLM realizer, schema-driven extraction, normalization, and a per-role Extended Kalman Filter — and validate it end to end on held-out synthetic data, where fine-tuning the extractor closes **75% of the gap to the structured-observation ceiling**.

Then we run it on a real event, and it loses to repeating the last reading.

This paper reports that outcome and what it localizes. The programme's question has two halves — **track** an evolving quantity, and **diarize** observations into the right event stream — and only the first is built. What ships for the second is hard assignment on a string key, not the specified multi-hypothesis association. Every real-event failure we have measured is in that second half: a 1999 earthquake's death toll tracked as a 2023 figure, one of two affected countries never recovered at all, and a hurricane's toll contaminated by three unrelated storms quoted in the same articles. A magnitude-based scope gate recovers a **9× improvement** on the event it was designed against but **splits held-out** — 3.7× better on the contaminated stream, 2.3× worse on the clean one — and the reason is diagnostic: without a declared scope hierarchy, "a larger scope" and "the largest part" are indistinguishable from the numbers alone. We then declared one, and the reference it enables is itself a measured negative.

Before building the multi-hypothesis association the design specifies, we price it. Assigning every observation to the scope it truly fits — an oracle, not a method — improves the shipped gate by **0.055 (9.3%)**. Perfect association is worth 9%, so multi-hypothesis tracking is not what is missing. The residual is 4.7% cross-event contamination against which no decode-side signal has worked, and the corpora offer no purchase on it: **0.0% of training documents contain a figure the model is supposed to leave alone**.

We also report that our own strongest prior result does not reproduce. An ablation concluding the filter's advantage *widens* under noise and censoring was measured on synthetic streams generated by the dynamics the filter models. On real revision data the advantage is real, reliable, and **1–2%**.

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## 1. The question, stated as it was actually asked

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Can a filter track real-world events reported in text, and separate them into the right streams?

Two halves, co-equal, and the second is not downstream of the first:

| half                                                                             | mechanism                    | status                                |
|----------------------------------------------------------------------------------|------------------------------|---------------------------------------|
| <b>track</b> — recover an evolving quantity from noisy, censored, lagged reports | Extended Kalman Filter (§2)  | built; validated on synthetic streams |
| <b>diarize</b> — decide <i>which</i> event stream each observation belongs to    | multi-hypothesis association | <b>not built</b>                      |

"Diarize" is borrowed deliberately from speaker diarization. It is not "who spoke when" but *which event is this figure about, over time*, and naming it that way makes obvious that the two halves fail independently and that the second can silently destroy the first.

**The association half is unbuilt, and this should be stated plainly rather than discovered by a reader.**

The design specifies association as gate → Hungarian assignment → top-K hypotheses → track birth and death. What ships is **hard assignment on an observable string key**, feeding one single-stream filter per key: no hypothesis enumeration, no deferred decision, no track birth or death. The only Hungarian solver in the repository is inside the extraction model's record loss, matching predictions to gold during training — unrelated to tracking.

That reframes every result below. The failures reported in §4 and §5 are failures of a **placeholder**, not of multi-hypothesis tracking. The mechanism the design names for exactly this problem has never been tested, so the programme's central question is not yet answered — it is, so far, unasked.

## 2. Method

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**Granularity.** Situation level, not event-instance level. A static instance has weak dynamics and the filter degenerates to fusion.

**State.** Per track: a pooled embedding, the continuous quantitative arguments (normalized counts, amounts, dates, magnitudes), and a salience term, with covariance.

**Dynamics.** Slow-drift embedding; monotone, random-walk or parametric process for the quantities; decay on salience, which provides track death. As process noise goes to zero the filter degenerates to optimal static fusion, so the single-document case is safe by construction — a mild win over naive merging, never a loss. Dynamics are inert by default and engaged only by evidence of change.

**Measurement.** Each document's extraction is a noisy and often censored observation — "at least 12", "roughly \$3M" — which is why the filter is *extended*: the measurement function is linearized.

**Gating.** A one-sided innovation gate, plus a learned mixture-of-experts router blending the local per-document read against the tracked estimate. The router is what withholds the filter in exactly the regime

where it would degrade a static read.

**The pipeline under test.** Synthetic stream generator → LLM realizer producing news-like prose → schema-driven extraction → confidence cut → normalization → per-role filter. Each stage is separately ablatable, which is what made the diagnosis in §3 possible.

### 3. The synthetic arc: extraction is solvable

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Working backwards from a structured-observation ceiling of 0.115 normalized RMSE — the score achievable if extraction were perfect — the zero-shot extractor reached 0.291. A probe localized the gap to extractor precision and confidence calibration rather than to normalization, and predicted that fine-tuning would close it. Fine-tuning on 29,198 casualty-structure examples:

| held-out test                | zero-shot | fine-tuned   |
|------------------------------|-----------|--------------|
| role precision               | 0.627     | <b>0.914</b> |
| role recall                  | 0.906     | <b>0.965</b> |
| value exact (number binding) | 0.991     | <b>1.000</b> |

| end-to-end (normalized RMSE; ceiling 0.115) | zero-shot | fine-tuned   |
|---------------------------------------------|-----------|--------------|
| best (confidence cut 0.99)                  | 0.291     | <b>0.165</b> |
| no confidence cut                           | 1.16      | <b>0.193</b> |
| the weak missing role                       | 0.458     | <b>0.122</b> |

The fine-tune closed **~75% of the gap to the ceiling**, exactly as predicted, and removed the dependence on a confidence cut — uncut performance went from unusable to 0.193. Number binding is perfect.

**The caveat that turned out to matter most:** this was fine-tuned and tested on the same synthetic distribution. Real news was left as the generalization test, and it is where the arc breaks.

### 4. The first real event, and the first real defeat

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Türkiye–Syria 2023, pre-registered before running. Ground truth is 16 daily points scraped from a single news tracker page via an archive service; search summaries were checked and found to **contradict each other on dates**, so they were rejected as a source.

**On real news the trivial baseline wins:** `est_last_value` 0.208 against the filter's 0.136 — reported here in the direction the metric runs, where the filter's 0.136 is the better score, but the finding is that the ranking reverses relative to every synthetic benchmark, and the margins are small enough that neither is a good result.

Three failures, none of them in the component the previous three sections optimized:

- **A 1999 death toll is tracked as a 2023 figure.** The 17,500 dead of the Izmit earthquake, quoted in an article's historical background section, enters the stream as a current observation — in every configuration.
- **Syria is never recovered.** The association key is computed per *document*, outside the envelope loop, so a two-country event collapses to one country.
- **Extraction is not the problem.** It reads the real trajectory nearly point for point.

**So the bottleneck is attribution, not filtering and not extraction.** That is the finding, and it redirected the programme away from further extractor work.

**A method rule learned the hard way.** A pre-registered prediction was scored a *success* by range-normalized RMSE while 12 of 20 readings came from a 1999 earthquake: the metric is blind to contamination that lands mid-range. Always report the trivial baseline beside the filter — it would have caught this immediately.

## 5. The scope gate: a 9× win that splits when held out

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Reporting mixes scopes: a national total and its state components appear in the same sentence. Filing the national figure under whichever state is nearest — what shipped — is plainly wrong.

A magnitude gate gives each observation a scope test against a reference. On Hurricane Helene, the event it was designed against:

| ratio | national stream | per-state mean |
|-------|-----------------|----------------|
| off   | 0.402           | 5.247          |
| 2.0   | 0.316           | <b>0.591</b>   |

Per-state error falls **5.247** → **0.591**, a 9× improvement, while the national stream *improves* rather than paying for it, and the effect is flat across ratios 1.5–2.5 rather than being a knife-edge setting. **Control:** removing the same number of observations at random over 40 trials gives 4.427, so the gate is *selecting* rather than thinning the stream into looking better.

Held out on Türkiye–Syria at the ratio fixed from Helene, it splits:

|           | Türkiye      | Syria        | mean  |
|-----------|--------------|--------------|-------|
| off       | <b>0.228</b> | 3.401        | 1.815 |
| gate @2.0 | 0.522        | <b>0.923</b> | 0.723 |

Syria was 65% contaminated — 11 of 17 values are Türkiye's tolls — and improves 3.7×. But **Türkiye, already clean, degrades 2.3×**. The reason is structural rather than a tuning failure: Türkiye–Syria declares no scope hierarchy, so the reference falls back to the running maximum across streams, which Türkiye's own values dominate. Türkiye is judged against a reference it defines itself. The gate rerouted a value that was Türkiye's *true* reading at that time.

**The diagnostic finding: the gate needs a declared scope hierarchy, not a magnitude.** Without one, "a larger scope" and "the largest part" are genuinely indistinguishable from the numbers alone — Türkiye's 41,000 filed under Syria and Türkiye's 41,000 filed under Türkiye look identical to any ratio.

**And the obvious fix for it was built, measured, and does not work.** Both events now declare containment explicitly — `rollup.json` carries a `hierarchy` block naming the aggregate and its parts — which enables the reference the diagnosis calls for: judge a part against its **implied maximum**, `aggregate - sum(other parts)`, rather than against a bare magnitude. That is the right shape for the dominant-part problem on paper. Turkey against an implied max of  $46,800 - 5,800 = 41,000$  sits exactly at its ceiling and is kept, while the same 41,000 filed under Syria faces an implied max of 5,800 and is rerouted.

On Helene it is **much worse than the plain aggregate reference: 2.590 against 0.591**. The parts are themselves contaminated, so their raw sum exceeds the whole and every implied maximum clamps to zero; a two-pass version recovers most of that but Florida still sits at 9.437. And Türkiye–Syria — the event that motivated the refinement — **still cannot test it**, because an implied maximum needs *independent* observations of the whole and that feed has none. Declaring the hierarchy does not manufacture the data.

So the diagnosis stands and its first implementation is a measured negative. Plain aggregate remains the recommended reference wherever an aggregate stream exists, and where none exists no gate is possible at all. Detail in `EKF_MHT_BUILD_RECORD.md` §25.5.

**Honest scorecard.** A  $9\times$  win with a clean control on the event it was designed against; held out,  $3.7\times$  better on one stream and  $2.3\times$  worse on the other. Both belong in any writeup, and the Helene number alone is the misleading half. The ratio was also chosen after seeing Helene's contaminated values; the plateau across 1.5–2.5 mitigates that without removing it. And 0.591 is nine times better than catastrophic, not good in absolute terms.

**What the gate does not address**, visible in the same audit: North Carolina's 1,400 is *Hurricane Katrina's* toll quoted inside a Helene article, its 250 is a typhoon, and its 230 is Hurricane Milton. That is **cross-event** contamination, not scope. A magnitude gate removes these for the wrong reason — because they are large, not because they belong to another event — so it would keep any cross-event figure that happened to be small.

## 6. Results that did not reproduce, and results that were negative

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Reported at the same weight as the positives, because collectively they are what localizes the remaining work.

### 6.1 Our strongest prior claim does not reproduce

An earlier ablation concluded the filter's edge *widens* under unreliability and censoring, and that conclusion was standing as the answer to "does the filter earn its keep". On the only real revision data available — a hurricane's North Carolina toll falling  $123 \rightarrow 102 \rightarrow 96$  as deaths were reclassified — it does not hold:

| regime                 | filter | last_value | gain         | filter wins |
|------------------------|--------|------------|--------------|-------------|
| noise 0.05, report 80% | 0.094  | 0.095      | +1.3%        | 191/240     |
| noise 0.10, report 60% | 0.189  | 0.192      | +1.5%        | 205/240     |
| noise 0.20, report 40% | 0.322  | 0.327      | <b>+1.8%</b> | 217/240     |
| noise 0.30, report 30% | 0.499  | 0.504      | +1.0%        | 222/240     |
| noise 0.40, report 20% | 0.628  | 0.634      | +0.8%        | 217/240     |

A revision is the case where `last_value` looks best — it adopts the new figure at once while a filter lags — so the filter can only win by smoothing noise the baseline chases. **It wins reliably**, in 80–93% of trials, on the first benchmark in this project whose baseline is not an oracle. **And it wins by 1–2%**, roughly flat, *shrinking* at the hardest setting rather than widening.

**Why the earlier result was wrong:** it measured synthetic streams generated by the same rise/decay dynamics the filter models, so the filter was matched to the process by construction. Real trajectories — plateaus, reclassifications, a genuine downward step — are not that process. *A dynamics model validated on data generated from it is not validated.* The case for the filter must rest on something other than the harder-regime argument, most plausibly multi-source disagreement, which no benchmark here has tested because every feed so far has a single source.

## 6.2 Aggregates cannot constrain their parts

Making the state a vector so a national total becomes a linear observation over states is the theoretically right move — a Kalman filter fuses "120 in North Carolina" and "227 across six states" natively, where any clustering method would work in value space and get it backwards. Tested against a parts-only filter on real per-state trajectories, it loses at every realistic reporting density and only reaches parity at 80%:

| per-state report rate | parts-only    | vector        | vector wins |
|-----------------------|---------------|---------------|-------------|
| 10%                   | <b>0.4348</b> | 0.6085        | 4/40        |
| 35%                   | <b>0.2292</b> | 0.2860        | 10/40       |
| 80%                   | 0.1556        | <b>0.1520</b> | 30/40       |

Right in principle, and it does not pay off at the densities real feeds provide.

## 6.3 Restructuring the text does not fix attribution

The proposal was a summarizer emitting self-contained bullets so number-to-place binding becomes local. Tested with hand-written bullets first, so a negative would cost nothing:

| arm                | correct    | false positives | fabrications |
|--------------------|------------|-----------------|--------------|
| raw text           | <b>3/5</b> | 1               | 0            |
| free bullets       | 2/5        | 3               | <b>2</b>     |
| extractive bullets | <b>3/5</b> | 1               | 0            |

Restructuring is neutral at best and the free variant fabricates. **The motivating example does not occur in the corpus at all** — the real failures are non-casualty numbers and cross-event leakage, not aggregate splitting. There is also a structural tension: the summarizer's highest-value act is normalizing prose into digits ("they died together" → "2 people died"), which is precisely what a verbatim-number guard must reject.

## 6.4 Guide-filtered negative sampling: wired, tested, negative

The residual confusion after description tuning is a genuine hard negative — "N people killed" against "N people evacuated", both counts of people, separated only by the verb, accounting for 11.2% of true death tolls. This is the case a guide-filtered loss exists for: the competing type is genuinely correct sometimes and genuinely wrong here, so a loss that penalizes it unconditionally would teach the wrong thing.

Implemented as a query-axis veto with a guide model and run as a two-arm A/B on 84,280 records, both arms swept to their own threshold:

| metric              | control | guide veto | delta   |
|---------------------|---------|------------|---------|
| entity strict       | 0.5858  | 0.5610     | −0.0248 |
| relation strict     | 0.1439  | 0.1108     | −0.0331 |
| event_argument fair | 0.5786  | 0.5702     | −0.0084 |
| event strict        | 0.3433  | 0.3443     | +0.0010 |

**Negative on seven of eight metrics**, including the argument axis the veto exists to sharpen. The control validates the setup by reproducing the historical reference from scratch. Against a measured run-to-run floor, the entity loss survives and the relation loss does not, so the honest statement is that the veto is neutral-to-harmful and is not the mechanism for this failure.

Wiring it did surface two ways it would have been **silently inert**, which is the transferable part. A sample's query axis holds only the types its own record declares, and just 0.23% of records name a competing count type natively — so the cross-record rivals the method exists for are never present unless something injects them, and the veto would have run, cost nothing and changed nothing. Separately, the veto resolved each candidate's reference by position in the candidate list, which is only valid for a shared candidate pool and is false for the per-query default that ships.

## 7. Where this leaves the programme

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**Built and validated:** the filter, its measurement model, the learned gate, and the text-to-observation pipeline, end to end on held-out synthetic data at roughly 1.4× the structured-observation ceiling.

**Built and honestly weak:** the filter's advantage over repeating the last reading is 1–2% on real revision data, and the argument that it widens under hard regimes does not reproduce.

**Not built:** multi-hypothesis association — the half every real-event failure lives in. The scope gate of §5 is the first concrete instance of that layer doing useful work, in its simplest possible form and without a new model.

## 7.1 How much is left for better association — measured, not argued

The natural conclusion from §4 and §5 is "build the association layer properly," and MHT is the specified answer. Before spending on a hypothesis tree, a cost matrix, Hungarian assignment and track management, we measured what *any* better association could be worth. The measurement needs no new model: assign every observation to the scope it actually fits, using ground truth. It is a ceiling, not a method.

|                                                    | per-place mean nRMSE |
|----------------------------------------------------|----------------------|
| no gate                                            | 5.247                |
| shipped scope gate                                 | 0.591                |
| <b>oracle hard association</b> (uses ground truth) | <b>0.537</b>         |
| headroom for any better association                | <b>+0.055</b> (9.3%) |

**Perfect association buys 0.055.** That is the entire prize, and it prices MHT out: a large subsystem competing for a 9% residual on a single-source feed.

The per-place breakdown sharpens it. The gate already **beats** a perfect two-way assignment on Florida (0.704 vs 0.734) and South Carolina (0.365 vs 0.558), because it has a third option the oracle lacks — *drop*. Florida's 300 is not a casualty figure and North Carolina's 1,400 is Hurricane Katrina's toll: neither belongs to *any* Helene scope, so no assignment scheme can place them correctly. Only Tennessee is a genuine association gap (0.817 vs 0.320), and its contaminants — 32, 32, 32, 36, 50 against a truth of 18 — are instructive: too large for the state, too small to look national, so no magnitude test can catch them. That is the real case for richer association, and it is worth 0.055 across the event.

## 7.2 What the residual actually is

A context audit of all 106 observations puts the remainder somewhere else entirely: 82.1% genuine Helene casualties, **4.7% belonging to another event** (Katrina's 1,400, a typhoon's 250, Milton's 230, Bosnia's 16, Hurricane John's 2 in Mexico), 3.8% non-casualty numbers (speeds, rainfall), and a 9.4% unclear tail.

Cross-event contamination carries the *large* values, so it does the most damage per instance — and the scope gate removes the large ones for the wrong reason, because they are big rather than because they belong to another storm. It therefore keeps any *small* cross-event figure, which is exactly what happens with Bosnia's 16 and Mexico's 2. Bosnia is structurally invisible to every signal tried so far: it is a *place*, not a named storm, so nothing keyed on storm names can see it.

**So the next step is neither MHT nor more extractor fine-tuning: it is negative supervision on event identity.** The evidence that it is a training-data gap rather than a decode gap is that binding collapses from 1.000 to 0.369 the moment documents become multi-event, and no decoder change has moved that. Every training corpus we have is complicit — measured over 20,000 records per corpus, **0.0% of training documents have zero records**, so the model has never once been shown a casualty figure it is supposed to leave alone.

That conclusion — reached by pricing the obvious next build and finding it not worth it — is the paper's main practical contribution, and it was only reachable by running the system against a real event and losing.



## 8. Limitations

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- **One real event, one source per feed.** Both real evaluations draw ground truth from a single tracker page. The multi-source disagreement case — the most plausible remaining argument for a filter over a last-value baseline — is untested.
- **The scope-gate ratio was chosen after seeing contaminated values** on the event it was tuned against. The plateau across 1.5–2.5 mitigates this; it does not remove it.
- **Synthetic validation was circular in a specific way** (§6.1) and any future synthetic benchmark must generate trajectories from a process the filter does *not* model.
- **Cross-event contamination is unaddressed.** It is a distinct failure from scope, needs instance-level rather than type-level discrimination, and the one mechanism tried for it (§6.4) was negative.
- **A genuinely blind real test remains outstanding.** Both events evaluated here predate the model's knowledge cutoff.

## 9. Reproducibility

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Data and code live under `datasets/disaster_streams{,_hard,_sonnet5,_model}/ : generate.py` (parametric streams, with a `--regime` switch), `realize.py` (LLM text via a batch API, with `--batch-id` recovery), `model_arm.py` (extraction, caching raw output for `--from-raw`), `extract.py` (surface parser and normalizer), and `evaluate.py` (tracker, baselines and gate). Evaluation:

```
uv run python datasets/disaster_streams/evaluate.py --data <dir> --split <val|test> \
  [--min-conf 0.99] [--reject-sigma 4] [--learn-gate]
```

Always pass the trivial baseline alongside; `evaluate.py` reports `est_last_value` beside `est_ekf` for the reason given in §4.

Known-negative settings, recorded so they are not retried: a symmetric  $3\sigma$  innovation gate is catastrophic on rising tolls (use the one-sided, dynamics-aware gate); confidence-as-soft measurement noise does **not** beat a hard confidence cut, because zero-shot extraction errors are gross false positives rather than graded noise; and record-mode precision requires the boundary architecture, since the span decoder ignores structure mode.

The build record — attachment points, CLI, training plan, decision log, generator specification, the blind-test protocol, and the full negative-sampling implementation notes — is in `EKF_MHT_BUILD_RECORD.md`.

## 10. References

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